

HIMANSHU BHARTI

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SUMMARY

Software Development Engineer specialized in production RAG systems and high-concurrency backend architectures. Expert in LLM orchestration (Gemini, OpenAI), retrieval optimization using BGE embeddings, and scaling AI-agent handoff workflows. Proven success in reducing query latency from ~1.2s to <300ms and deploying multi-tenant AI SaaS platforms with strict multi-tenant isolation.

TECHNICAL SKILLS

Languages: Python, Java, SQL, TypeScript, JavaScript

AI Systems: RAG Pipelines, FAISS, LangChain, BGE-M3/MiniLM Embeddings, Semantic Chunking, Cross-Encoder Reranking

Backend: FastAPI, Spring Boot, RESTful APIs, Distributed Locking (Redis), WebSockets, JWT, RBAC

Infrastructure: AWS (EC2, S3), GCP (Vertex AI), Docker, Kubernetes, PostgreSQL, MongoDB, Vector Databases, CI/CD

AI SYSTEMS EXPERIENCE

- **Retrieval Optimization:** Reduced query latency from ~1.2s to <300ms by implementing FAISS indexing and Redis-level caching.
- **RAG Performance:** Improved answer relevance by 40% using Hybrid Search (BM25 + Vector) and metadata-aware retrieval.
- **Agentic Workflows:** Designed logic-based "Agent Handoff" triggers monitoring conversation friction to escalate queries via WebSockets.

PROFESSIONAL EXPERIENCE

Software Development Engineer (Full-Stack & AI) May 2025 – Present
Mobiloitte Technologies New Delhi, India

- Deployed omnichannel AI bots (WhatsApp, Instagram, Slack) for 10k+ users with domain-specific knowledge bases.
- Designed a SaaS-based HITL system for AI-to-agent transitions, reducing support ticket turnaround time by ~50%.
- Implemented semantic chunking for multi-tenant PDF parsing, ensuring 0% data leakage between client vector stores.
- Optimized multi-tenant billing and LLM usage logging within administrative dashboards using secure JWT-based RBAC.

Full Stack Developer Intern Aug 2024 – Apr 2025
QSpiders Noida, India

- Built high-integrity transactional systems using Java Spring Boot and MySQL, managing concurrent data updates and ACID compliance.
- Optimized SQL indexing and complex execution plans, resulting in a 20% increase in average API throughput and reduced query time.

SELECTED PROJECTS

AI-Powered Social Media Automation Platform | *Java, Spring Boot, Vertex AI, Redis, Capacitor*

- Developed a high-concurrency generation engine using Gemini for multi-format (Posts, Reels, Threads) content creation at scale.
- Implemented Redis distributed locking to prevent double-billing in credit-based user accounts during high-traffic spikes.
- Deployed a unified React + Capacitor codebase for web/mobile, reducing maintenance and development overhead by 40%.

University Management System & AI Chatbot | *Python, FastAPI, Next.js, MongoDB, AWS, Docker*

- Integrated a voice bot using WebSockets for real-time admission inquiries and campus visit scheduling via STT/TTS pipelines.
- Architected containerized backend systems on AWS for Admissions and HR, ensuring 99.9% uptime and enterprise scalability.
- Built a RAG-based Chatbot automating 60% of inquiries via semantic search on unstructured campus documents and PDF archives.

Enterprise AI-Powered Business Platform | *Python, Next.js, FastAPI, Redis, LangChain*

- Developed a multi-format RAG pipeline for context-aware queries against complex PDF and Excel datasets.
- Implemented automated PII masking and audit logging to enforce strict data privacy and isolation for multi-tenant clients.

EDUCATION

Bihar Engineering University | B.Tech in Electronics & Communication Engineering | CGPA: 7.8/10